

Cecil Ash

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Summary

- 5+ years owning end-to-end data platforms and ML systems at high-growth SaaS companies.
- Architect and lead developer of 3 data/ML products launched within 6 months.
- M.A.Sc. and B.Sc. Industrial Engineering, Dalhousie University | [Peer-reviewed publications](#)
Researched applications of stochastic optimization models to supply chain decision-making.

Work Experience

Machine Learning Engineer II • Teamworks • USA

2023–Present

Data Platform & Generative AI • *Tech: AWS, Databricks, Onehouse, Postgres, PySpark, Kafka*

- Productionized a new lakehouse platform that unified fragmented vendor and product data from acquired companies into a shared foundation for AI-enabled products.
- Identified and validated AI use cases for the data platform, driving adoption of Claude Code agents for pipeline and infrastructure development, debugging, and log monitoring.
- Developed PySpark pipelines within a medallion architecture to transform raw data into production-ready datasets powering LLM queries and generative AI applications.

Event-Driven ML Infrastructure • *Tech: GCP, Python, DuckDB, dbt, Kubernetes, Docker, Celery, Datadog, Terraform*

- Built a new ML pipeline for on-demand inference, including model training and deployment on GPU-enabled clusters, serving APIs, and production monitoring and alerting in Datadog.
- Implemented Kubernetes cost monitoring in GCP by standardizing job labels and correlating billing + usage data into dashboards and alerts, reducing response time to cost spikes.

Data Ingestion & Entity Resolution API • *Tech: GCP, Python, Kubernetes, Docker, Airflow, Terraform*

- Launched a new data integration product that unified customer data from multiple vendors.
- Researched and prototyped fuzzy ID mapping techniques, reducing manual entity-matching effort.
- Improved ETL runtimes 5× through optimized data caching and partitioning.
- Mentored junior engineers and provided workstream oversight, improving delivery quality.

Data Engineer • Karbon • Canada

2022–2023

Multi-Tenant Analytics & BI Platform • *Tech: Snowflake, Python, dbt, Power BI, Elasticsearch, Terraform*

- Built a production multi-tenant data platform and Snowflake warehouse from scratch, spanning API ingestion, dbt models, Terraform infrastructure, and Power BI.
- Led development of an [award-winning](#) business intelligence product generating \$500K ARR.
- Designed dimensional data models (star schema) to support scalable BI and reporting.
- Built data ingestion pipelines for accounting APIs, handling rate limits, retries, and schema evolution.
- Reduced dbt runtimes by 75% through SQL optimization and partitioning strategies.

Analytics Engineer • Modern Data Science Inc. • Canada

2021

- Transformed and analyzed usage data to identify expansion and churn signals for SaaS companies.

Side Projects

Dual-RAG Evidence Retrieval and Writing-Style Adaptation

2026

- Built a dual-RAG application that retrieves resume evidence and writing-style examples to generate tailored application materials, with a feedback loop that learns from user-accepted edits.

Cost-Native Multi-Agent Coding Harness (Rust)

2026

- Built a multi-agent Rust harness with real-time token/USD cost tracking and budget control loops that enforce spend ceilings and manage graceful degradation across agent fleets.